

17. $\begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 5 \\ -7 \\ 4 \end{bmatrix}, \begin{bmatrix} 6 \\ 3 \\ 5 \end{bmatrix}$

Solution: Let $A = \begin{bmatrix} 0 & 5 & 6 \\ 1 & -7 & 3 \\ -2 & 4 & 5 \end{bmatrix}$. By the Invertible Matrix Theorem and the equivalence of the statements (d) and (e),

we know that the equation $Ax=0$ has only the trivial solution if and only if the columns of A form a linearly independent set.

$Ax=0$. By Theorem 3, this has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.

$$\begin{array}{c}
 \begin{bmatrix} 0 & 5 & 6 & 0 \\ 1 & -7 & 3 & 0 \\ -2 & 4 & 5 & 0 \end{bmatrix} \xrightarrow[\text{Row 1 and 2}]{\text{Interchange}} \begin{bmatrix} 1 & -7 & 3 & 0 \\ 0 & 5 & 6 & 0 \\ -2 & 4 & 5 & 0 \end{bmatrix} \\
 \begin{bmatrix} 1 & -7 & 3 & 0 \\ 0 & 5 & 6 & 0 \\ 0 & -10 & 11 & 0 \end{bmatrix} \xrightarrow[\text{+2xRow2}]{\text{Row3=Row3}} \begin{bmatrix} 1 & -7 & 3 & 0 \\ 0 & 5 & 6 & 0 \\ 0 & 0 & 23 & 0 \end{bmatrix} \xrightarrow[\text{x1/23}]{\text{Row3=Row3}} \begin{bmatrix} 1 & -7 & 3 & 0 \\ 0 & 5 & 6 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\
 \begin{bmatrix} 1 & -7 & 3 & 0 \\ 0 & 5 & 6 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[\text{-3xRow3}]{\text{Row1=Row1}} \begin{bmatrix} 1 & -7 & 0 & 0 \\ 0 & 5 & 6 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[\text{-6xRow3}]{\text{Row2=Row2}} \begin{bmatrix} 1 & -7 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\
 \begin{bmatrix} 1 & -7 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[\text{+7xRow2}]{\text{Row1=Row1}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}
 \end{array}$$

It's easy to verify that the final matrix is in rref. Each of the 3 columns of A has a pivot. $\therefore$ The system has no free variables. By Theorem 2 from Chapter 1, the system of equations has a unique solution. $\therefore$ The columns of A are linearly independent. By the Invertible Matrix Theorem and the equivalence of the statements (e) and (h), we know that the columns of A span $\mathbb{R}^3$. $\therefore$ The given vectors form a basis for $\mathbb{R}^3$.